

# CHARLES (CHUCK) FORSYTH

Research Computing and AI Executive | HPC | Research Cloud | Navy Veteran

Los Angeles, CA | [chuck.forsyth@gmail.com](mailto:chuck.forsyth@gmail.com)

[linkedin.com/in/chuckforsyth](https://linkedin.com/in/chuckforsyth) | [charles-forsyth.github.io](https://charles-forsyth.github.io) | [github.com/charles-forsyth](https://github.com/charles-forsyth) | ORCID 0000-0003-4872-0296

## EXECUTIVE SUMMARY

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Research computing and AI executive with 30+ years in technology, from Navy radar to high-performance computing to generative AI. As Director of Research Computing at UC Riverside, I lead a matrix team of about 20 across ITS and the colleges and run the cloud, cluster, storage, security and AI services behind about 1,000 researchers in 22 colleges and schools, and I still build the platforms myself, now with a team of AI agents. I led UCR's first-of-its-kind research partnership with Google Cloud, served as Co-PI on a \$498K NSF CC\* research storage award, and chaired the UC Research IT Committee across every UC campus. I have been building with AI since the GPT-2 era, running GPU jobs for it on the National Research Platform in 2020, and shipped an LLM assistant for HPC users in 2023. The number I manage to is time to science.

## CAREER HIGHLIGHTS

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- **Google Cloud partnership:** led Ursa Major, announced by Google in 2023 as a first-of-its-kind research subscription; 170+ research projects under one governed, budgeted cloud.
- **NSF award:** Co-PI on CC\* award 2346636 (\$498,553), which delivered 2.09 PB of research storage on the 100 Gbps Science DMZ, 422 TB shared with the National Research Platform.
- **AI for research:** built UCR's governed AI stack (MCP servers, AI Gateway, elastic GPU cluster) and ship it with AI agents: 240+ releases across ten tools in five weeks.
- **Campus and systemwide leadership:** founded and lead UCR Research IT Leaders; Research Advisory Board and Academic Senate research committee; Chair of the UC Research IT Committee (2022-2024), uniting research IT leaders from every UC campus.

## CORE COMPETENCIES

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- **Leadership:** AI and research computing strategy, governance and policy; budgets and FinOps; NSF grants; vendor and industry partnerships; systemwide committees
- **AI:** agentic AI and MCP servers, governed model gateways (Gemini, Claude, open-weight), LLM tools for researchers, AI security and audit
- **Infrastructure:** HPC and Slurm/PBS, Google Cloud and AWS, Kubernetes, Ceph, Lustre, GPFS, InfiniBand, Linux; Python, Go, Bash; Postgres and pgvector
- **Security:** NIST 800-171, CMMC, P3/P4 data security plans, secure research enclaves

## PROFESSIONAL EXPERIENCE

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### Director of Research Computing

Mar 2023 - Present

*University of California, Riverside, Riverside, CA*

Lead research computing strategy, services and AI for about 1,000 researchers in 22 colleges and schools through a matrix team of about 20: direct reports and student staff, the ITS systems team (dotted line for research) and research IT staff in colleges, centers and units across campus.

- Run Ursa Major, the Google Cloud research partnership: per-project budgets, automated spend throttles, security baselines and FinOps across 170+ projects.
- Co-PI of NSF CC\* 2346636 (CephRDS, 2.09 PB usable); six science driver labs migrated, 422 TB federated to the National Research Platform.
- Built the AI-for-research stack: MCP servers with per-user Google sign-in, roles and a full audit trail; the UCR AI Gateway; and an elastic Slurm cluster on Google Cloud (CPU, high-memory, NVIDIA L4 GPU) with zero compute cost when idle.
- Built Nexus, the system of record Research Computing runs on: 943 researchers, 431 org units, 176 cloud projects, 12,000+ links.
- Data security plans and secure enclaves for P3/P4 and CMMC research with the ISO.
- Founded and lead UCR Research IT Leaders, which coordinates research IT across campus; serve on the Research Advisory Board and the Academic Senate research committee.
- Official UCR PI status (2024); grant strategy; ship software with a team of AI agents under written specs, test gates and human approval.

### Associate Director of Research Computing

Aug 2019 - Mar 2023

*University of California, Riverside, Riverside, CA*

- Developed research computing grant strategy and managed budgets for research technology initiatives within Information Technology Solutions (ITS).

- Lead systems architect for research computation, advanced statistical computing and network interoperability with the campus Science DMZ.
- Proxy for the CIO and CTO in academic senate technology discussions; supervised technical staff and set fiscal, HR and resource allocation policy.
- Brought UCR researchers onto National Research Platform GPUs: TensorFlow, PyTorch and GPT-2 jobs (2019-2020).

### Chair, UC Research IT Committee

Jul 2022 - Jan 2024

*University of California (systemwide), Remote*

Vice Chair, Jun 2021 - Jul 2022.

- Led the UC Research IT Committee, liaising with the UC IT Leadership Council (ITLC) and the UC Office of the President CIO.
- United research IT leaders from every UC campus around shared priorities; set and met annual goals and advocated systemwide for UC researchers' computing needs.

### Lead Linux & HPC Systems Administrator / Consultant

Jun 2017 - Aug 2019

*University of California, Riverside, Riverside, CA*

- Architected, installed, administered and supported UCR's campus HPC systems; built and ran on-demand HPC clusters in AWS.
- XSEDE Campus Champion: connected UCR researchers and students to NSF supercomputing centers and national laboratories.
- UCR representative to the UC Research IT Committee and to CaRCC.

### Lead Linux & HPC Systems Administrator / Research Consultant

Jan 2012 - Jun 2017

*Corning Incorporated, Corning, NY*

Lead HPC architect and administrator for Corning's research, development and engineering computing.

- Linux clusters with thousands of cores, Lustre, InfiniBand, SAN and NAS storage; 60+ network-licensed science codes (Ansys, COMSOL, Abaqus).
- 400+ hardware support contracts and 150 kVA+ of data center power and cooling; led internal and external projects and wrote training and documentation.

### Earlier Career

- **Owner and Research System Administrator, CnL Networks**, Jun 2003 - Jan 2012. Founded and ran a systems administration and IT outsourcing business; concurrently Technical Account Manager, STCR (Apr 2008 - Aug 2009).
- **Linux System Administrator, Hancock Telephone Company**, Mar 1998 - Jun 2003.
- **Hardware Test Engineer, IBM**, Aug 1997 - Mar 1998; Technical Support, AOL, Mar - Aug 1997.
- **Advanced Electronics Technician, United States Navy**, Dec 1994 - Mar 1997. Radar, communications and IFF systems aboard the Aegis cruiser USS Lake Champlain (CG-57).

### AI AND RESEARCH PLATFORMS I BUILT

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- **ursa-bifrost**: MCP server that lets any AI assistant work a Slurm cluster through typed, audited tools (Go, MIT, [github.com/UCR-Research-Computing/ursa-bifrost](https://github.com/UCR-Research-Computing/ursa-bifrost)).
- **UCR AI Gateway**: one governed endpoint to Gemini, Claude and open models with per-lab budgets and a FinOps dashboard.
- **Nexus**: research relationship knowledge graph and OAuth MCP server with 41 role-gated tools.
- **deep-research**: Gemini Deep Research workstation that checks report claims as real HPC jobs (MIT, [github.com/charles-forsyth/deep-research](https://github.com/charles-forsyth/deep-research)).
- **Ultra AI Workstation**: AI operator desk for mail, tickets, Slack and the cluster, with two-step approval on AI drafts (MIT).
- **Skywalker**: audit engine for hundreds of Google Cloud projects.
- **Ursa Major Ask (2023), uma (2024), HPC Job Gallery (2026)**: early LLM tools for HPC users; 49 tested Slurm workflows.

### PROFESSIONAL SERVICE AND LEADERSHIP

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- UCR Campus Champion, NSF XSEDE and ACCESS (2017-present); NSF grant peer reviewer (2020-2021)
- Co-Chair, UCR Visualization Lab Steering Committee; UCR representative to CaRCC, CASC and ACI-REF; research collaborator, Pacific Research Platform (2017-present)

### PUBLICATIONS AND TEACHING

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- "One Day on the Research Stack," Field Notes, Oct 2026. [charles-forsyth.github.io/papers/](https://charles-forsyth.github.io/papers/)
- The AI + HPC Field Manual, 15 chapters, 2026. [charles-forsyth.github.io/handbook/](https://charles-forsyth.github.io/handbook/)
- Taught Linux, Python and AWS HPC clusters at UCR (2017-2019); NRP Nautilus researcher guide, 8 articles (2026)

### EDUCATION AND CERTIFICATIONS

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- B.S., Information Technology, Purdue University Global, 2015 (GPA 3.89); Networking and Technical Support Specialist, Ridley-Lowell Institute, 2011 (GPA 4.0); Advanced Electronics Technology, U.S. Navy, 1997.
- **Certifications**: PBS Professional (Altair), Mellanox InfiniBand, EMC Isilon, Kubernetes, PMI Project Management Foundations.